AUREOLYZE

Attributions

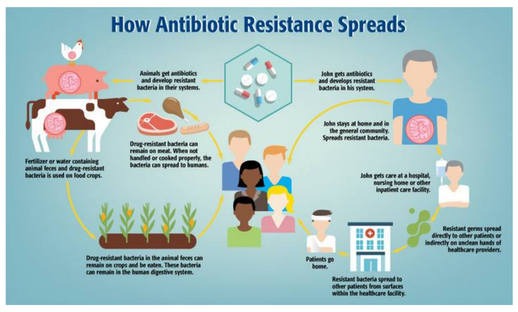
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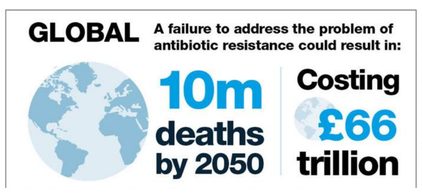
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Sponsors

Breaking the Biofilm Barrier: Combating Resistance with Synthetic Biology







Biofilms are highly organized and resilient microbial communities encased within a self-produced extracellular matrix that anchors bacterial cells to surfaces and shields them from hostile environments. This matrix serves as both a physical and biochemical barrier, protecting embedded bacteria from immune clearance, desiccation, and, most importantly, antibiotic treatment.

PROBLEM STATEMENT

- Naturally occurring ESP is known to **degrade a few human proteins such as ECM proteins, il33, c5 complement system, etc**. and most importantly ***S.Aureus* Autolysin and biofilm proteins.** Its toxicity towards humans hinders its ability to act as a therapeutic agent.
- It comes under the class of glutamyl serine endopeptidases. Its overall structure and mechanism of cleavage resemble chymotrypsin but it cleaves after glutamyl residues located interior to proteins

OUR SOLUTION

We propose engineering Extracellular Serine Protease (ESP) produced by *S.epidermidis*, for targeted *S.Aureus* biofilm degradation with minimal human ECM impact. ESP offers a novel approach to combat biofilm formation and existing biofilms on orthodontic devices. Through site-directed mutagenesis (SDM) and computational analysis, we aim to modify specific amino acid residues that **determine the substrate specificity of the enzyme**. By fine-tuning active site interactions in ESP, we seek to **retain its biofilm-cleaving efficiency while minimizing off-target effects on host tissues.**

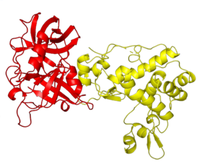
Challenges Faced by Current Technologies

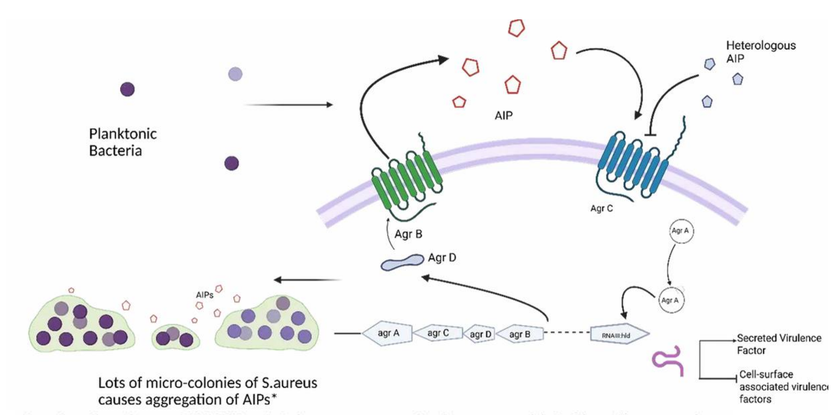
- **Poor Penetration: Antibiotics cannot breach the dense EPS(Extracellular Polymeric Substances) matrix**
- **High dosage of antibiotics evolutionarily aking biofilms more robust and unaffected**
- **Treatment varies for different bacteria, i.e., there is no universal tretament**
- **Silver Nano Particles damage the epithelial cells and interfere with DNA replication**

Inspiration from Nature

Extracellular Serine Protease

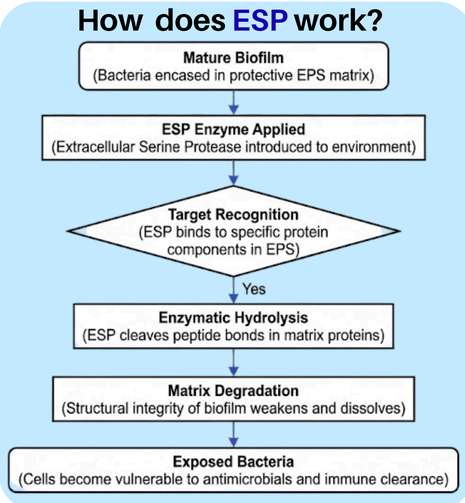
ESP, an enzyme capable of targeted protein hydrolysis. **Cleaves the biofilm matrix** and plays a major role in infection control, **dissolving the bacterial shield to allow effective antibiotic penetration**





AgrA stimulates RNAIII, which manages virulence and biofilm dispersal. Low bacterial densities stick to surfaces; high bacterial densities produce toxins for invasion. Reduced Agr activity in mutants correlates with increased chronic biofilm formation.

How does ESP work?

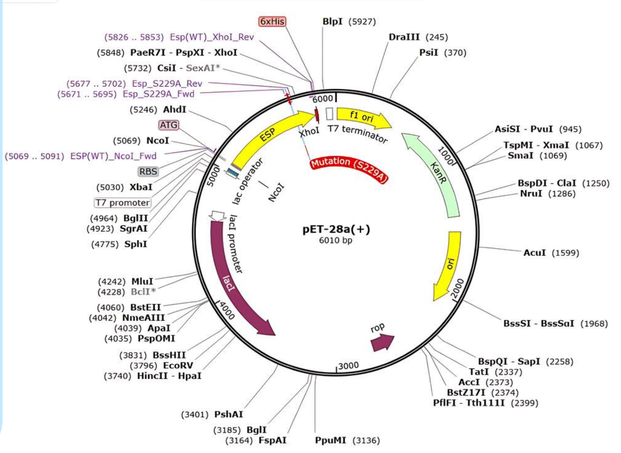


WHAT IS AUTOLYSIN?

While exploring proteins critical for *S.Aureus* biofilm development, the autolysin (Atl) stood out as a central factor. Atl is the major peptidoglycan hydrolase in *S.Aureus* and plays a dual role:

- **Cell Wall Remodelling & DNA Release:** Atl cleaves the bacterial cell wall during cell division and autolysis. This controlled lysis releases extracellular DNA (eDNA), a key structural component of the biofilm matrix. Without Atl-mediated DNA release, *S.Aureus* struggles to form robust biofilms.
- **Adhesion & Biofilm Initiation:** Beyond lysis, Atl fragments function as adhesins, promoting bacterial attachment to host proteins (fibronectin, fibrinogen) and abiotic surfaces. This makes Atl critical during the early stages of biofilm establishment.

PLASMID MAP



Experiment Design-1

Expression of Wild Type-ESP(WT-ESP) from *S. Epidermidis* in *E.Coli*

- Synthesized the sequence of the gene coding for **ESP- Backbone: pET-28a**
- Heterologous gene sequence would contain a flag-tagged sequence downstream that will help in purification, specifically its a His-tag
- The protein would be expressed in E.coli expression cell lines **(BL21, DE3).** The protein is expressed with induction using IPTG The expression and extraction of WT-ESP was done for the later control for the variants of the ESP.
- Use of biophysical techniques like Circular dichroism (CD) and Bradford's method.

Experiment Design-2

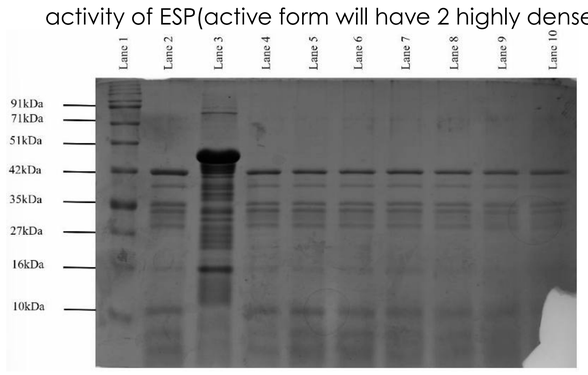
- Creation of variants of WT-ESP by **Site Directed Mutagenesis**
- Designed the different primers to create multiple variants of Wild Type by changing the amino acids of few sites
- Sites changed: **V67A, S229A, T230A, N256A.**
- Created two fragments of the gene sequence by using set of primers ordered.
- These two fragments of genes were stitched together using **Stitch PCR.**
- To confirm the formation of the stitched gene, Gel electrophoresis was used, presence of single band signified the success of the process

Experiment Design-3

- **Transformation** of the variants in **DH10β** and expression
- DH10β is a high-efficiency cloning strain of E. coli well-suited for cloning large plasmids, making it an ideal host for our mutant constructs
- Using Ca shock method to induce transformation The protein would be **expressed in E.coli expression cell lines (BL21, DE3).**
- The protein is expressed with induction using IPTG Purification was done by Nickel Affinity, Ion exchange and Gel Chromatography.
- DH10β would act as our primary source for the variants of ESP for further experiments

Experiment Design-4

- The activation of pro-peptide ESP to mature peptide ESP via thermolysin cleavage
- After expression the protein we get is inactive and gets activated by the thermolysin
- Using different concentrations of Thermolysin to check the activity of ESP(active form will have 2 highly dense streaks)



Cleavage of pro-peptide ESP purified from chromatography was treated with thermolysin to convert it to the mature peptide, which is the active form. Lane 1: Ladder, Lane 2: Protein, Lane 3: Thermolysin, Lane 4: R1, Lane 5: R2, Lane 6: R3, Lane 7: R4, Lane 8: R5, Lane 9: R6, Lane 10: R7

MODELLING OF BIOFILM FORMATION

- Biofilm formation is inherently a **spatiotemporal process,** i.e., it depends on both space and time, cells attach to a surface, multiply, and secrete an extracellular matrix that thickens over time. Most experimental studies observe biofilm growth in terms of surface coverage or total biomass. Still, these **fail to capture** how **biofilm thickness evolves** spatially due to **nutrient gradients** and **diffusion limitations.**
- To overcome this limitation, we developed a spatial **biofilm–substratum model** that explicitly represents the vertical structure of the biofilm. This approach enables us to investigate how nutrient diffusion, biomass accumulation, and substrate consumption change with depth and time, providing a mechanistic understanding of biofilm expansion.
- This model is a mechanistic framework for describing biofilm growth on a degradable substratum, adapted from the **Wanner–Gujer formulation.** It couples a diffusion–reaction Partial Differential Equations for nutrient concentration within the biofilm to Ordinary Differential Equations governing biofilm thickness and substratum thickness

$$\frac{dP}{dt} = \mu_p P - k_{\text{attach}} P + (k_{\text{detach,QS}} + k_{\text{detach,prot}}) B - d_p P$$

$$\frac{dB}{dt} = k_{\text{attach}} P + \mu_b B - (k_{\text{detach,QS}} + k_{\text{detach,prot}}) B - d_{b,\text{tot}} B$$

$$\frac{dS}{dt} = -\frac{1}{Y}(\mu_p P + \mu_b B)$$

$$\frac{dE}{dt} = \alpha_E B - \gamma_E E - v_{\text{EPS}}$$

$$\frac{dA}{dt} = \alpha_A B - \delta_A A$$

$$J(\lambda) = \frac{Y D_c}{X} \left.\frac{dC}{dz}\right|_{z=\lambda}$$

$$g(C) = \begin{cases} 0, & C \le \tilde{c}, \\ k \cdot \dfrac{(C-\tilde{c})^2}{\kappa^2 + (C-\tilde{c})^2}, & C > \tilde{c}, \end{cases}$$

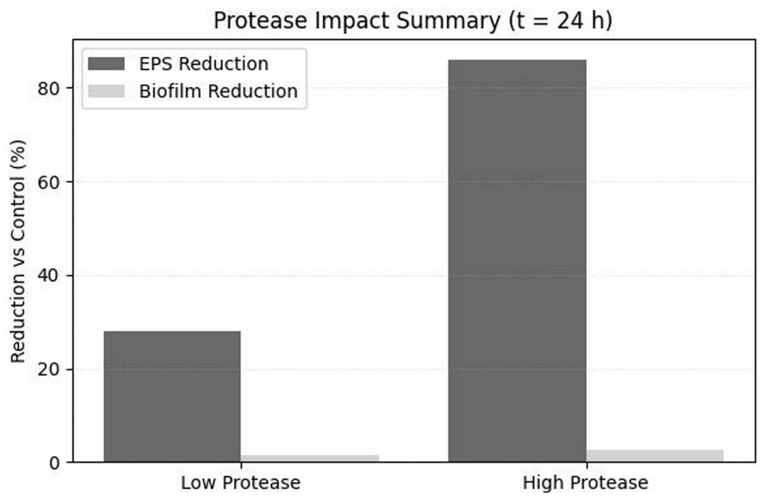
$$\frac{dS}{dt} = -\,g(C(0))\,\frac{S}{S+\varepsilon_S}\,\frac{\lambda}{\lambda+\varepsilon_\lambda}$$

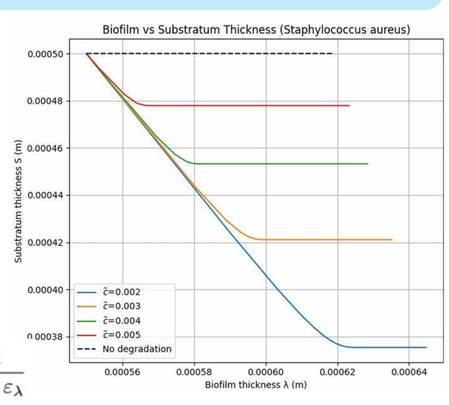
$$D_e \frac{d^2C}{dz^2} - \frac{\mu X}{Y}\frac{C}{K_s + C} = 0, \quad 0 < z < \lambda$$

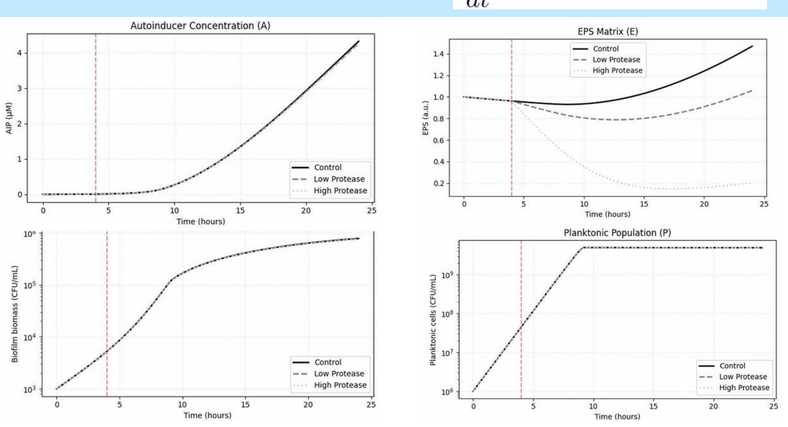
$$\frac{d\lambda}{dt} = J(\lambda) - (k_l + \delta)\lambda + A\,g(C(0))\,\frac{S}{S+\varepsilon_S}\,\frac{\lambda}{\lambda+\varepsilon_\lambda}$$

MODELLING OF BIOFILM DEGRADATION

We developed an ODE model incorporating a protease-mediated degradation mechanism that emulates the effect of the extracellular serine protease (ESP) from *S.Epidermidis*. The model simulates the dynamic interplay between planktonic cells, biofilm biomass, EPS, quorum sensing peptides (AIP), and substrate concentration under protease exposure. By applying Michaelis–Menten kinetics to describe enzyme-driven EPS breakdown, we quantitatively evaluate how engineered ESP variants targeting the autolysin Atl could accelerate biofilm disruption without affecting cell viability.



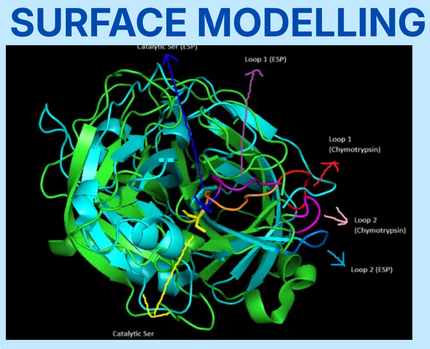




APPLICATION

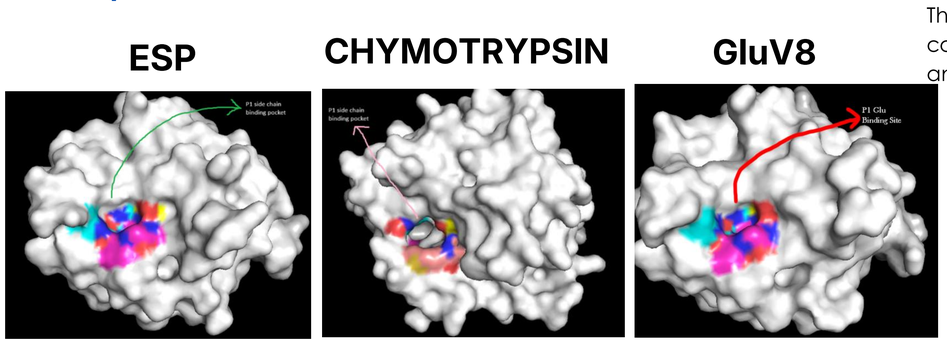
Aureolyze is an enzyme-based biofilm disruption platform that uses ESP protease, with optional DispersinB and low-dose antimicrobials, to break down biofilm matrices and **restore bacterial susceptibility without promoting resistance**. It is applied as a contact lens solution, catheter coatings and lock therapies for urinary and bloodstream infection prevention, a pH-responsive dental system for oral biofilm control, and a nebulized adjunct therapy for cystic fibrosis. Across all settings, Aureolyze integrates into existing routines and targets biofilms at their molecular core.

SURFACE MODELLING



- **3D** representation of the protein's surface, **highlighting pockets, grooves, and cavities** that may serve as substrate-binding sites or active sites. By examining the shape and accessibility of these binding pockets, surface modelling helps to infer how proteins recognise and process specific substrates.

we compared its structural features with those of well-characterised serine proteases such as chymotrypsin and GluV8



These findings suggest that while ESP, GluV8 & chymotrypsin share the conserved serine protease fold, subtle variations in loop architecture and pocket dimensions define their substrate preferences.

ESP	Chymotrypsin	GluV8
Cyan: Val 67, His 250, and Thr 230	Cyan: Gly 216 and Gly 226	Cyan: Val 1, His 184, and Thr 164
Yellow: Ser 235	Yellow: Ser 195	Yellow: Ser 169
Pink: Oxyanion hole (231 to 234)	Pink: Ser 189	Pink: Oxyanion hole (165 to 168)
	Peach: Oxyanion hole and surrounding residues (192 to 194)	

Team Aureolyze

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